4	**(a)**	**(i)**	$V = q / 4\pi\varepsilon_0 R$	B1	[1]
		(ii)	(capacitance is) ratio of charge and potential or q/V	M1	
			$C = q/V = 4\pi\varepsilon_0 R$	A0	[1]
	(b)	**(i)**	$C = 4\pi \times 8.85 \times 10^{-12} \times 0.45$	C1	
			$= 50$ pF	A1	[2]
		(ii)	either energy = ½ CV^2 or energy = ½ QV and $Q = CV$	C1	
			energy of spark = $\frac{1}{2} \times 50 \times 10^{-12} \{(9.0 \times 10^5)^2 - (3.6 \times 10^5)^2\}$	C1	
			$= 17$ J	A1	[3]